

Testing indicator strategies on NQ futures

RSI, ADX and daily range ideas, tested in-sample 2022-2024 and out-of-sample 2025-2026 | 1 October 2026

Bottom line. Four ideas were tested on NQ (five counting the inverted RSI version). Three had no edge, either in sample or out of sample. One idea, **fading the break of the previous day's range**, made money in both periods. But its out-of-sample profit comes almost entirely from a single trade on 9 April 2025, so it is not validated. Without that one trade, the out-of-sample result is about breakeven.

How the tests were run

- **Data:** NQ tick data, 2022 to September 2026. 5-minute bars were built from the ticks using the front-month contract each session, with prices adjusted across contract rolls so rolls don't create fake signals.
- **Split:** In-sample (IS) 2022-2024 for building and comparing variants. Out-of-sample (OOS) 2025-2026 was run *once*, only on a cell chosen in advance by a written rule, and never used to tune.
- **Lock rule (declared before looking):** the best profit-factor cell with enough trades (150 on 5m, 100 on daily), where every neighbouring cell on the grid is also profitable. If no cell qualifies, the idea fails and OOS is not run.
- **Fills and costs:** signals on bar close, filled at the next bar's open. If a stop and a target are both hit inside one bar, the stop wins. Cost is 1 point round trip (about \$20). All results are for 1 contract at \$20 per point.
- **Sharpe:** annualised, computed from daily P&L over all trading days.

Results at a glance

#	Idea	IS net	OOS net	Verdict
1	RSI crossing its EMA + volume (5m), and the inverted version	All 21 cells lose	Not run	No edge
2	ADX rising + volume, direction from price vs EMA(50) (5m)	+\$36.6k (Sharpe 0.54)	-\$11.3k (Sharpe -0.19)	Failed OOS
3	Daily: break of previous range in a strong trend, stop at previous extreme (also without the trend filter)	All cells lose (-\$32k to -\$61k)	Not run	No edge
4	Daily: fade the break of the previous range	+\$100.7k (Sharpe 0.97)	+\$55.5k (Sharpe 0.53)	Survives, but one trade dominates

1. RSI, RSI-EMA and volume on the 5-minute chart

Rules. RSI(14) crosses above its own 9-period EMA, RSI between 50 and 70, and volume at least 1.5x the average for that time of day: go long. Shorts mirror this (cross below, RSI 30 to 50). Stops tested: fixed 15, 20, 25 and 30 points, and ATR(14) multiples of 1x, 2x and 3x. Each at 1:1, 1:2 and 1:3 reward-to-risk. 21 cells in total.

Fixed stop	1:1	1:2	1:3	ATR stop	1:1	1:2	1:3
15 pts	0.63	0.83	0.87	1x ATR	0.99	0.89	0.87
20 pts	0.82	0.93	0.87	2x ATR	0.88	0.88	0.95
25 pts	0.91	0.98	0.91	3x ATR	0.91	0.90	0.82
30 pts	0.95	0.89	0.91				

In-sample profit factor, both directions (1.00 is breakeven). Every cell is below 1, with 330 to 450 trades per cell.

- With costs set to zero the results sit around 0.9 to 1.05, so this is not a cost problem. There is no underlying edge.
- **Inverted** (long when it said short, and vice versa): also no edge. Fixed stops lose in all 12 cells (profit factor 0.66 to 0.96). A small pocket with ATR 2x (profit factor 1.06 to 1.10) is surrounded by losing neighbours, so it fails the lock rule.
- Trade frequency is low: about 0.47 trades per day, on about a third of days. Signals cluster midday and afternoon.

2. ADX instead of RSI (5-minute)

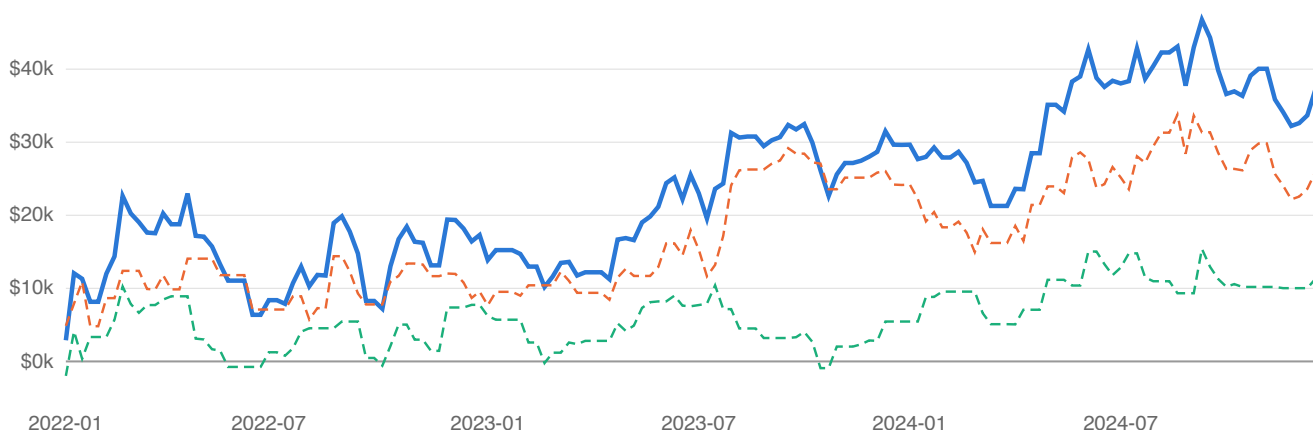
Rules. ADX(14) crosses above its own 9-period EMA (trend strength rising), with volume at least 1.5x normal. Direction was first taken from +DI vs -DI, then from price above or below its 50-period EMA (“trade with the EMA”). The two versions gave almost identical results.

In-sample, ATR stops formed a plateau: all 9 cells had a profit factor of at least 1.00. The best locked cell was **ATR 3x stop, 1:3 reward-to-risk**.

	Net P&L	Sharpe	Profit factor	Max drawdown	Trades
In-sample 2022-2024	+\$36,550	0.54	1.15	-\$19,075	342
Out-of-sample 2025-2026	-\$11,340	-0.19	0.95	-\$33,230	217

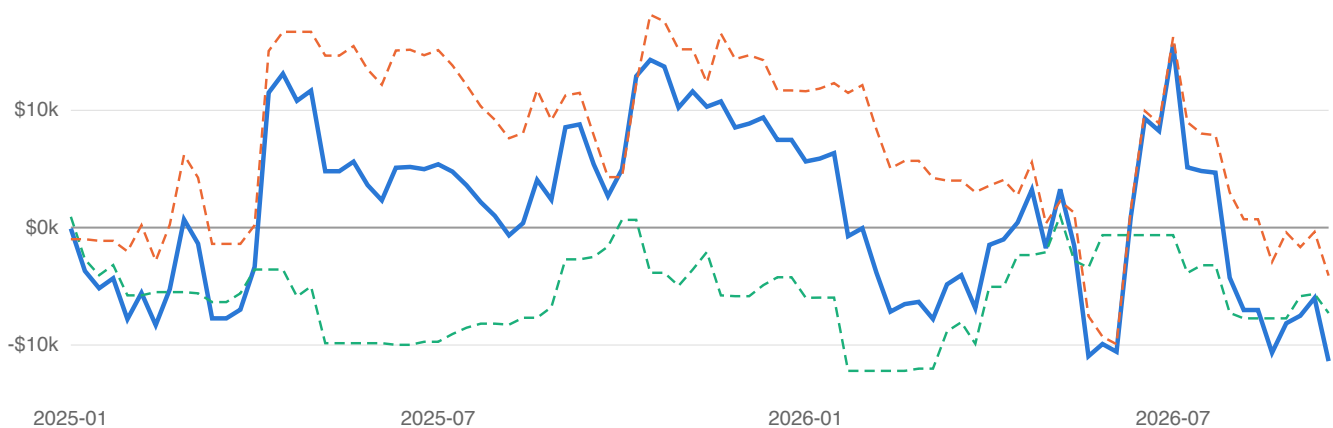
ADX 5m, in-sample 2022-2024 (ATR 3x stop, 1:3 RR)

Combined Long Short



ADX 5m, out-of-sample 2025-2026

■ Combined ■ Long ■ Short



- The “1:3” label is misleading. The stop is wide (median about 77 points), so the target is rarely reached: only 16 of 342 in-sample trades hit it. Most trades were closed at the end of the session.
- In-sample profit also shrank every year (2022 +\$17.3k, 2023 +\$12.3k, 2024 +\$6.9k). A Sharpe of 0.54 over three years is a t-statistic of only about 1, which is within what luck produces after trying many cells.

3. Daily breakout of the previous day's range

Rules. Daily bar = one futures session. If ADX(14) is at or above a threshold (20, 25 or 30) and ATR(14) is above its 50-day average, enter on a break of the previous day's high (long) or low (short), in the direction of +DI vs -DI. Stop at the previous day's opposite extreme. Exits tested: 1R, 2R, 3R targets and hold to the close.

- Every one of the 12 cells lost money in-sample: -\$32k to -\$46k, profit factor 0.73 to 0.86. A stricter trend filter made it worse.
- Removing the ADX filter did not help. Keeping direction from +DI/-DI: -\$42k to -\$57k. Trading whichever side breaks first, with no trend filter at all: -\$56k to -\$61k (profit factor about 0.9).
- The stop averages about 350 points (roughly \$7,000 of risk per contract) and is rarely hit. The target almost never fills, so every exit variant behaves like “hold to the close”.

OOS was not run, because no cell passed the lock rule.

4. Fading the break of the previous day's range (daily)

Rules. Same trigger and volatility filter (ATR(14) above its 50-day average), but trade *against* the first break: short if yesterday's high breaks first, long if yesterday's low breaks first. Because the old stop would sit on the profit side, a stop beyond the level of 0.25x, 0.5x or 1.0x of yesterday's range was tested, with 1:1 / 1:2 / 1:3 targets and hold to the close.

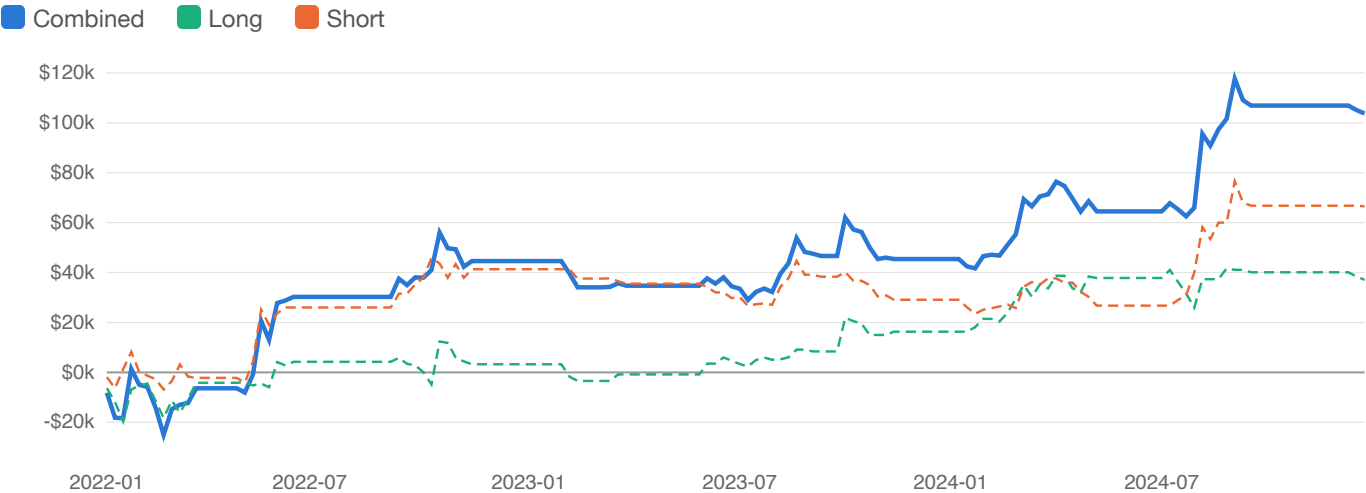
In-sample grid (both directions): net P&L, with Sharpe

Stop beyond level	1:1	1:2	1:3	Hold to close
0.25x range	+\$42,272 Sharpe 0.74	+\$58,389 0.76	+\$73,742 0.83	+\$100,726 0.97
0.5x range	+\$62,055 0.67	+\$87,842 0.78	+\$97,932 0.83	+\$93,042 0.79
1.0x range	+\$46,995 0.37	+\$41,180 0.31	+\$42,480 0.32	+\$42,480 0.32

All 12 cells are profitable (profit factor 1.08 to 1.30), a genuine plateau, and the long-only and short-only grids are positive in every cell too. The lock rule picked **0.25x range stop, hold to the close**. Making fills harder (entry must trade 1 to 2 points through the level, plus 1 to 2 points of extra slippage) still gave +\$85k to +\$94k in-sample.

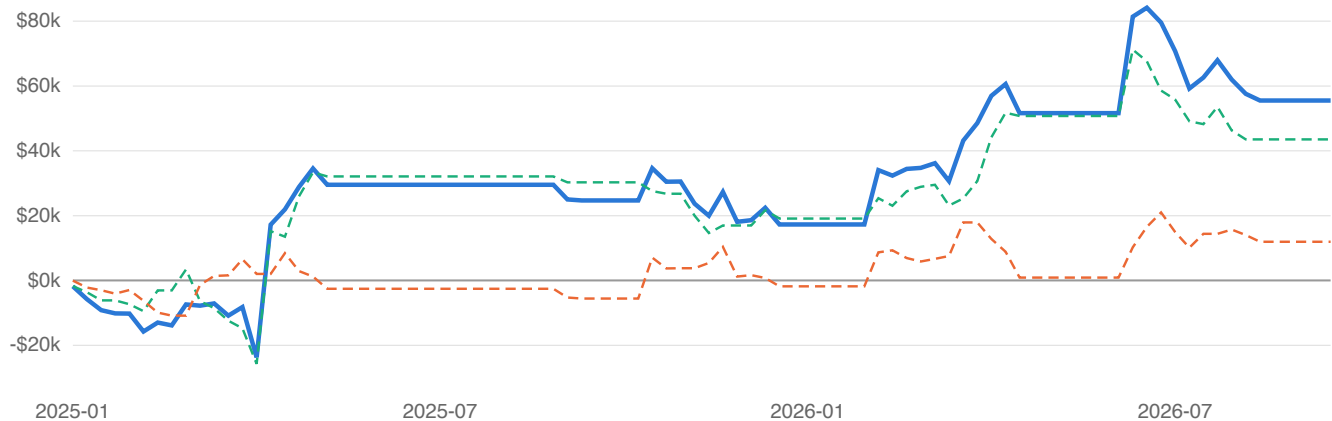
	Net P&L	Sharpe	Profit factor	Win rate	Max drawdown	Trades
In-sample 2022-2024	+\$100,726	0.97	1.30	33%	-\$27,015	330
Out-of-sample 2025-2026	+\$55,498	0.53	1.17	32%	-\$35,556	201

Daily fade, in-sample 2022-2024 (0.25x range stop, hold to close)



Daily fade, out-of-sample 2025-2026

■ Combined ■ Long ■ Short



Why this is not validated yet

- **One trade makes most of the OOS profit.** On 9 April 2025 (the tariff-pause rally, NQ from about 17,000 to 19,354), the strategy went long on a break of the prior low and made +2,379 points, +\$47,580. Without it, OOS net is about +\$7,900. Removing the five best trades turns +\$55.5k into -\$58.6k.
- **It is a tail-catcher.** About two out of three trades lose, and the money comes from rare days when a break turns into a violent reversal. In-sample, a third of the profit came from about six weeks in autumn 2024, and 2023 was flat (+\$0.8k).
- **Long dead stretches:** about a year flat in 2023, and 2025 sat below zero until the April spike, then gave back about half of the gain by year-end.
- **Risk size:** stops average about 87 points in-sample and 133 in OOS (\$1,700 to \$2,700 per contract), with a worst single trade of -\$6,600 OOS. That fits poorly with a prop firm's daily-loss or trailing-drawdown limits.
- **Selection:** this was the fourth idea (fifth counting the inverted RSI version) tried on the same data, and it was tested only after the breakout version failed. That counts against any single positive result.

What would come next

- Test whether a filter or sizing rule makes the fade less dependent on a handful of days (for example, skipping sessions after extreme-range days).
- Check the 2023 flat year and 2025 for what regime the fade does not work in, before adding any rules. Fewer rules is the goal.
- Re-run on a longer history (2021 and earlier) if available, since three years in-sample contains only a few tail events.

Research summary only. Past and simulated results do not predict future results and this is not investment advice. Backtests are on one contract with a simple cost model (1 point round trip, no market impact, stop-wins tie-breaking). Real fills, gaps and rollover days can differ. Data: NQ tick archive, bars built per session with roll adjustment; strategies simulated in standalone Python.